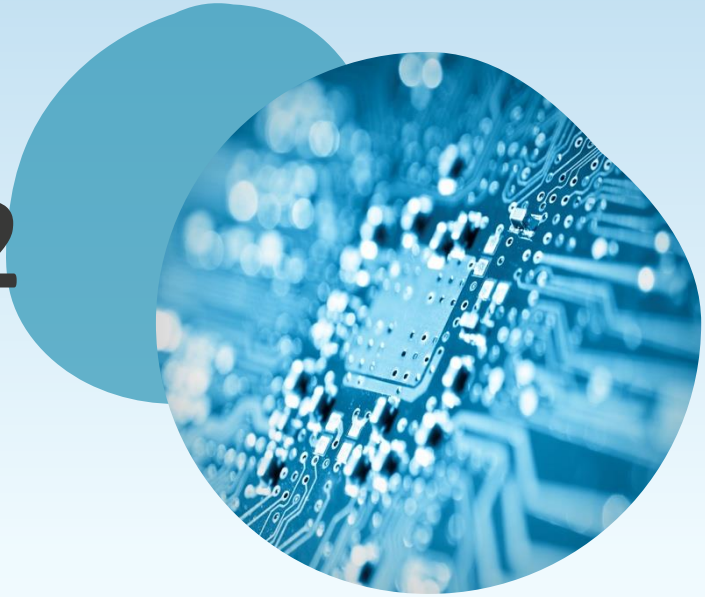
Interfacing MSP432 BOOSTERPACK SENSORS - II

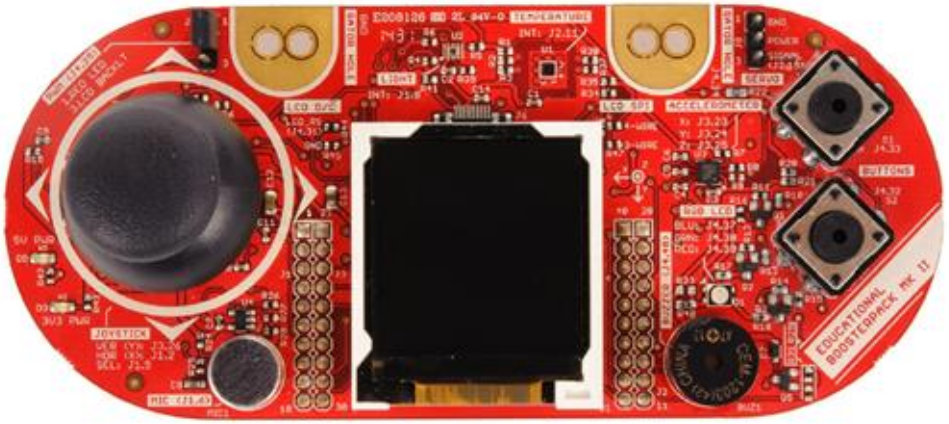


Embedded Software for the Internet Of Things

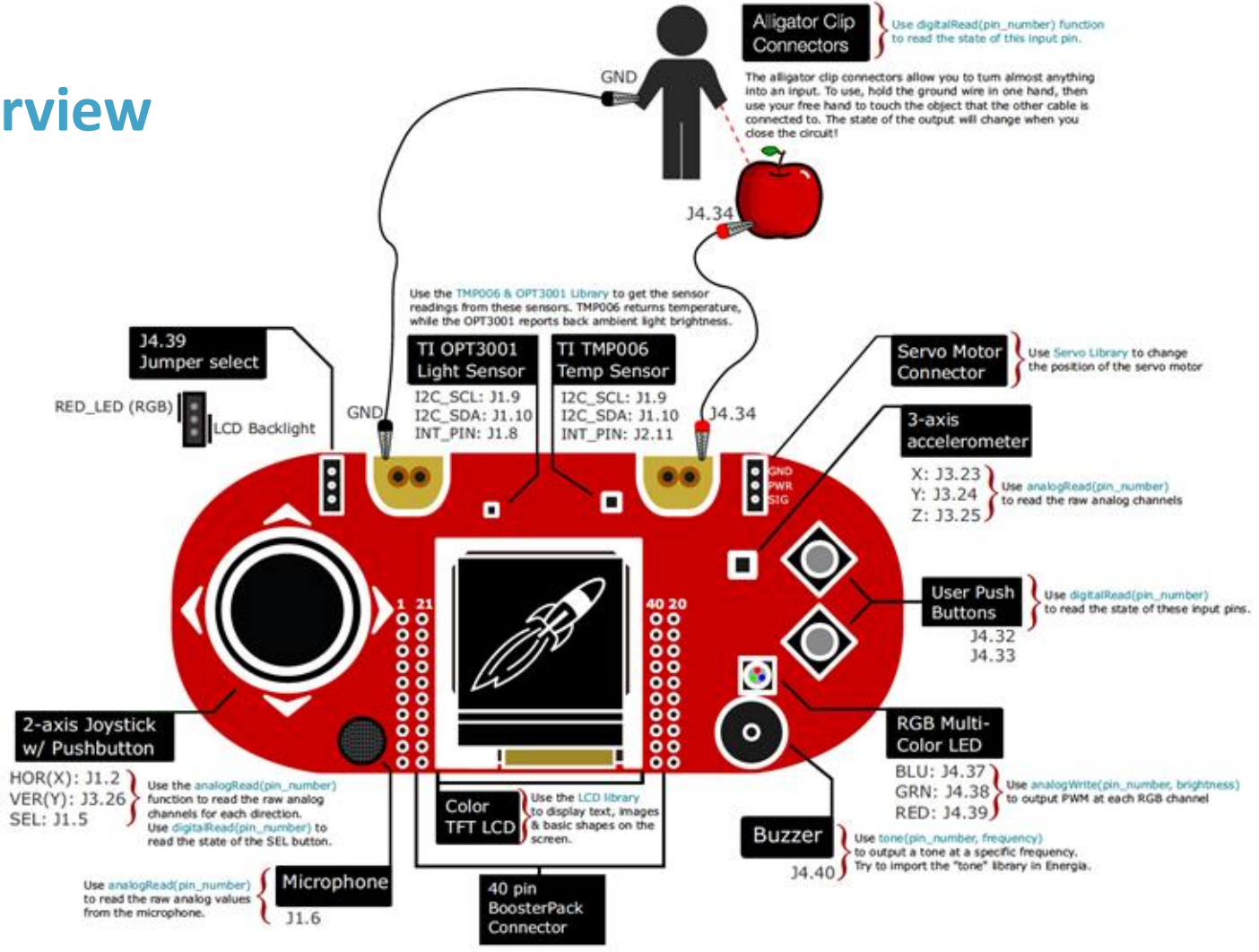
Kasim Sinan Yildirim

BOOSTXL-EDUMKII EDUCATIONAL BOOSTERPACK

- Various analog and digital inputs and outputs
- TI OPT3001 Light Sensor
- TI TMP006 Temperature Sensor
- Servo Motor Connector
- 3-Axis Accelerometer
- User Push Buttons
- RGB Multi-color LED
- Buzzer
- Color TFT LCD Display
- Microphone
- 2-Axis Joystick with Pushbutton



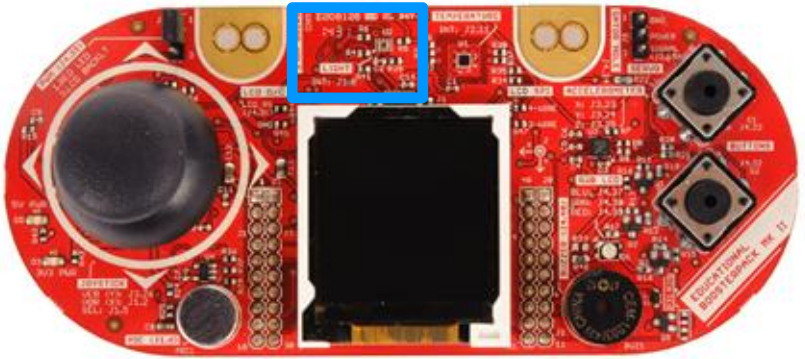
Sensor Overview



TI OPT3001 LIGHT SENSOR

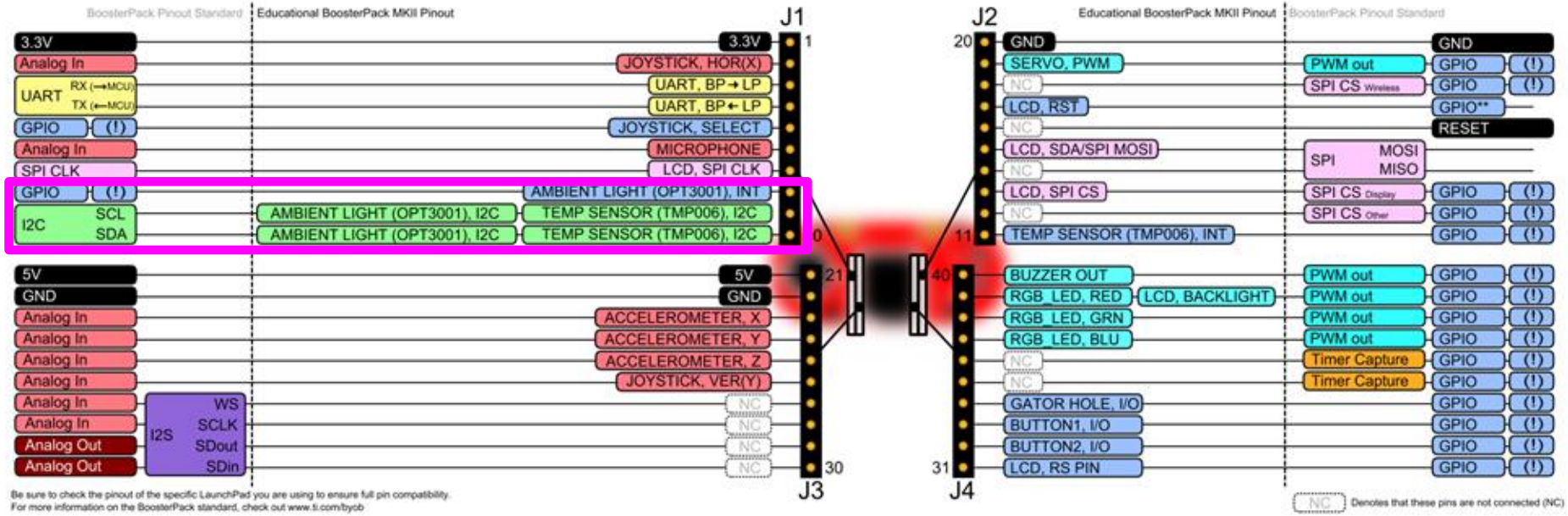
- The **OPT3001** is a digital **ambient light sensor (ALS)** that measures the intensity of light as visible by the human eye.
  - Covering the sensor with your finger or shining a flashlight on it will change the output of the OPT3001.

BoosterPack Plug-in Module Header Connection	Pin Function
J1.8	OPT3001 interrupt
J1.9(1)	I²C SCL
J1.10(1)	I²C SDA



PinOut Diagram

- Pins 6.5 (SCL) and 6.4 (SDA)



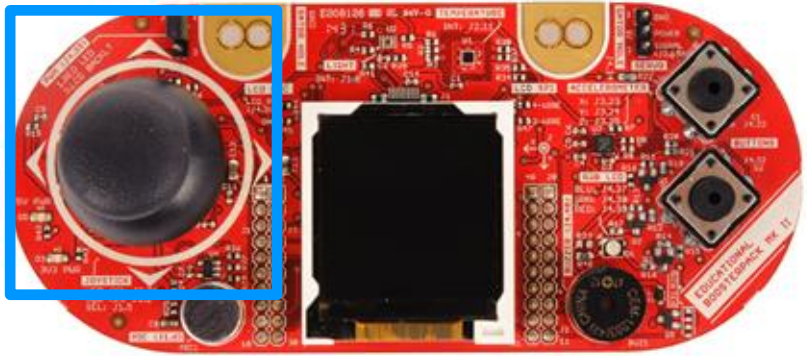
Lab#1: Occupancy Detection

- Try to implement “**occupancy detection**” using the Light Sensor.
  - If you detect occupancy, then toggle LED!

2-Axis Joystick with Push Button

- The ITEAD studio IM130330001 **2-axis joystick** with pushbutton is simply **two potentiometers**, one for each axis.
  - Read **the voltage** present on the joystick axis to provide the position of the joystick to the application
  - The select button is actuated when the joystick is pressed down.

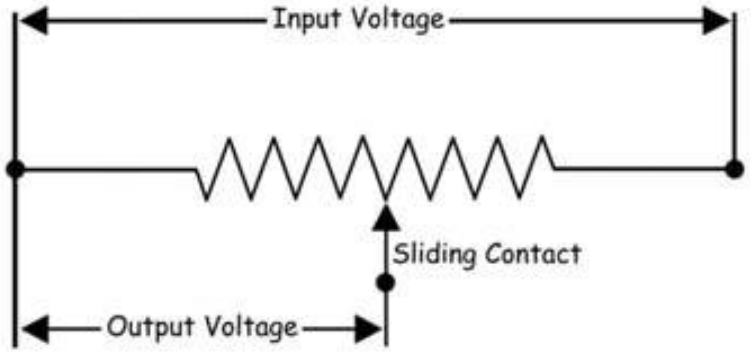
BoosterPack Plug-in Module Header Connection	Pin Function
J1.2	Horizontal X-axis
J1.5	Select button
J3.26	Vertical Y-axis



What is a Potentiometer?

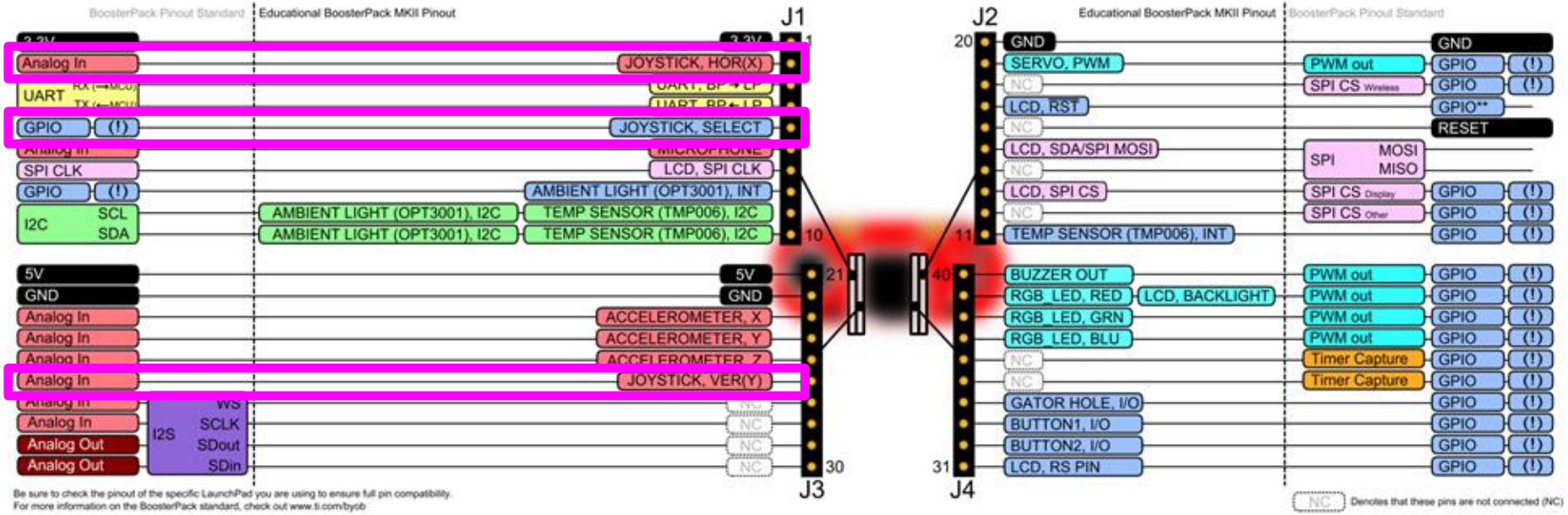
- It’s a resistor with a **variable resistance** by turning the ‘wiper’.
- Position can be read as a variable **analog voltage**.





PinOut Diagram

- Pins 6.0 and 4.4 on launchpad are X and Y axis.



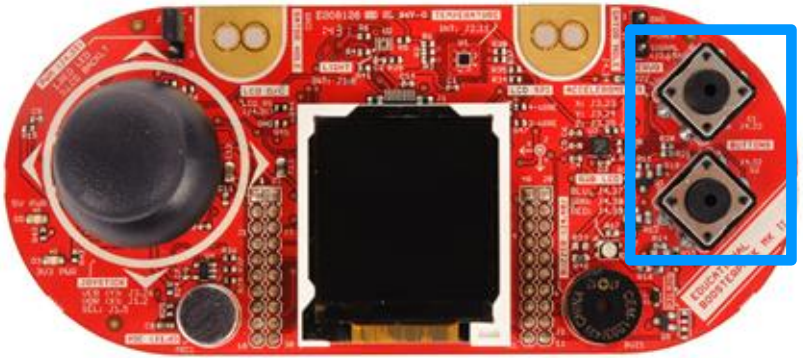
Lab #2: Moving Objects on the Screen

- Put a character on LED screen and shift it up/down and left/right based on the Joystick input.

User Push Buttons

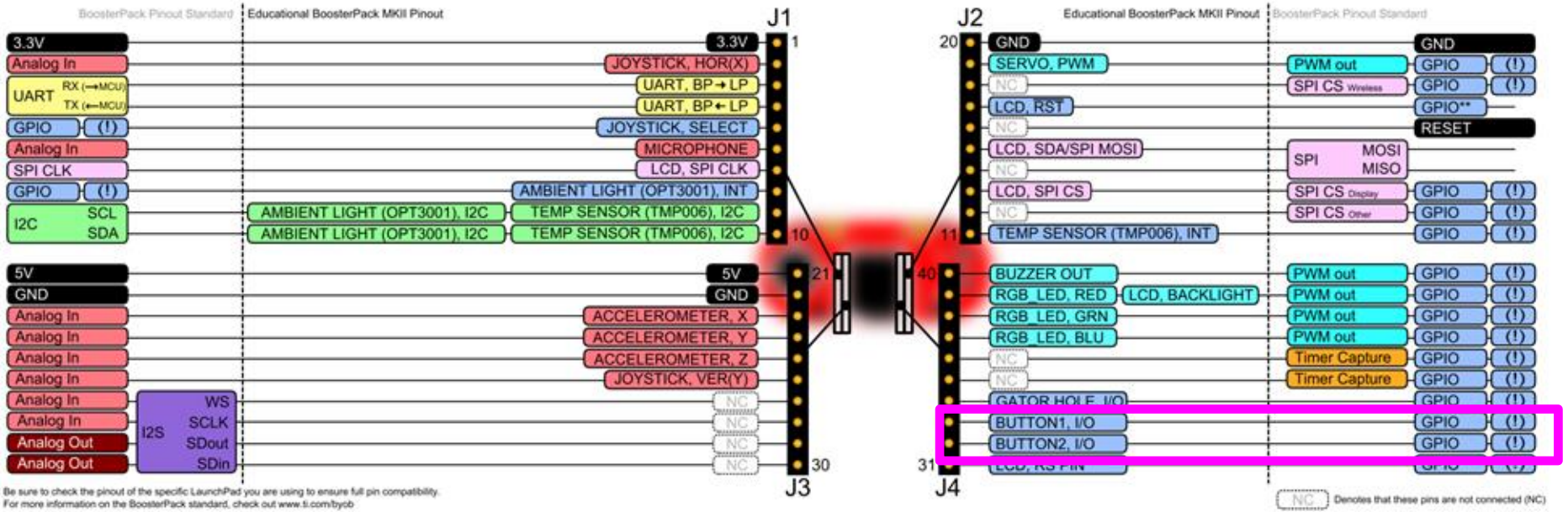
- The user pushbuttons on the BOOSTXL-EDUMKII **are connected to pullup resistors** that drive the BoosterPack plug-in module.
  - **Pin high** until the button is pressed and then the pin is driven **low**.

BoosterPack Plug-in Module Header Connection	Pin Function
J4.32	S2 button
J4.33	S1 button



PinOut Diagram

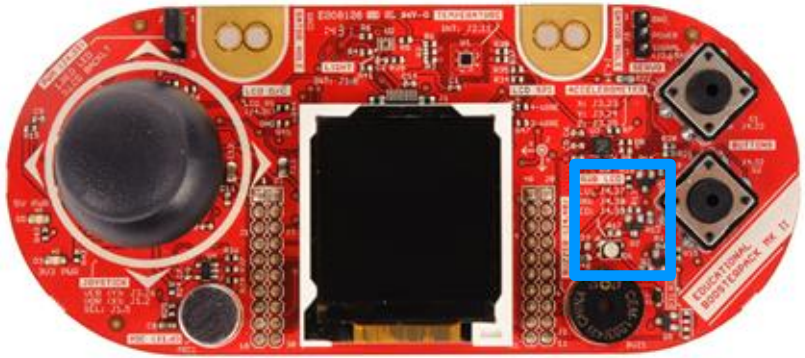
- Guess which pins?


BoosterPack Pinout Standard
Educational BoosterPack MKII Pinout
J1
J2
Educational BoosterPack MKII Pinout
BoosterPack Pinout Standard
3.3V
Analog In
UART
RX (→MCU)
TX (←MCU)
GPIO
(!)
Analog In
SPI CLK
GPIO
(!)
I2C
SCL
SDA
5V
GND
Analog In
Analog In
Analog In
Analog In
Analog In
Analog In
Analog Out
Analog Out
I2S
WS
SCLK
SDout
SDin
3.3V
JOYSTICK, HOR(X)
UART, BP→LP
UART, BP←LP
JOYSTICK, SELECT
MICROPHONE
LCD, SPI CLK
AMBIENT LIGHT (OPT3001), INT
AMBIENT LIGHT (OPT3001), I2C
TEMP SENSOR (TMP006), I2C
AMBIENT LIGHT (OPT3001), I2C
TEMP SENSOR (TMP006), I2C
5V
GND
ACCELEROMETER, X
ACCELEROMETER, Y
ACCELEROMETER, Z
JOYSTICK, VER(Y)
NC
NC
NC
NC
1
10
21
30
20
11
40
31
GND
SERVO, PWM
NC
LCD, RST
NC
LCD, SDA/SPI MOSI
NC
LCD, SPI CS
NC
TEMP SENSOR (TMP006), INT
BUZZER OUT
RGB_LED, RED
RGB_LED, GRN
RGB_LED, BLU
NC
NC
GATOR HOLE, I/O
BUTTON1, I/O
BUTTON2, I/O
LCD, RS PIN
LCD_BACKLIGHT
PWM out
SPI CS Wireless
GPIO
GPIO
GPIO**
RESET
SPI
MOSI
MISO
SPI CS Display
SPI CS Other
GPIO
GPIO
GPIO
PWM out
PWM out
PWM out
PWM out
Timer Capture
Timer Capture
GPIO
GPIO
GPIO

Multicolor LED Light

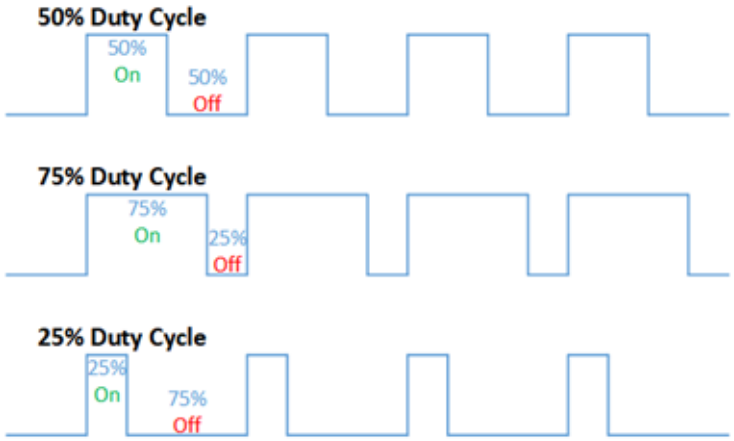
- The Cree CLV1A-FKB RGB multicolor LED light output can make any color **by mixing** red, green, and blue.
  - Each color channel can be individually modified by **pulse width modulation (PWM)** to achieve the desired color.

BoosterPack Plug-in Module Header Connection	Pin Function
J4.37	Blue channel
J4.38	Green channel
J4.39(1)	Red channel



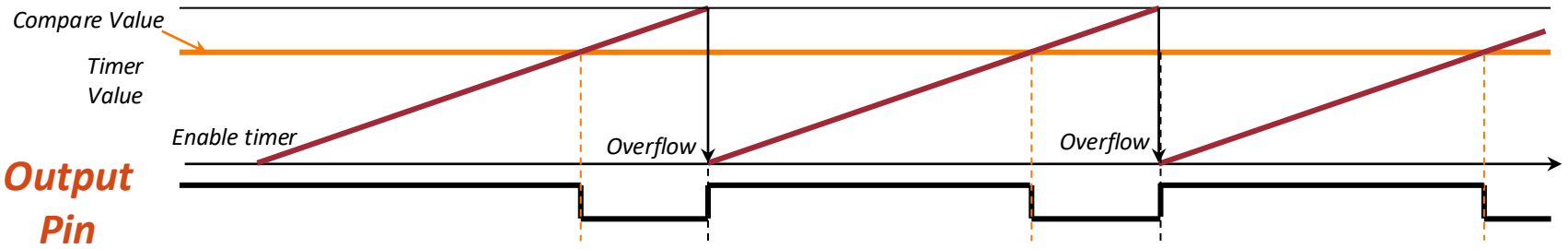
Pulse-Width Modulation

- **PWM** signal characteristics
  - **Modulation frequency:**
    - How many pulses occur per second
  - **On-time**
    - amount of time that each pulse is on
  - **Duty-cycle** = on-time/period
- Adjust on-time (hence duty cycle) **to represent the analog value**



Timer - PWM Mode

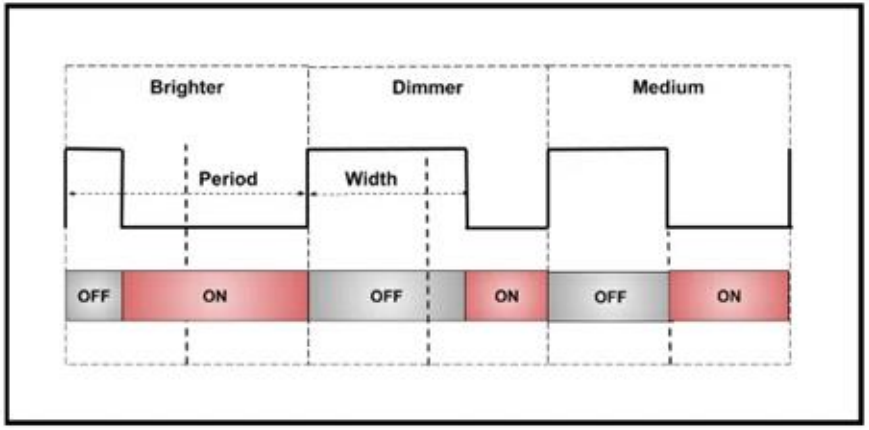
- PWM duty cycle is proportional to compare value
  - Period: Max timer value
  - Pulse Width: Compare Value

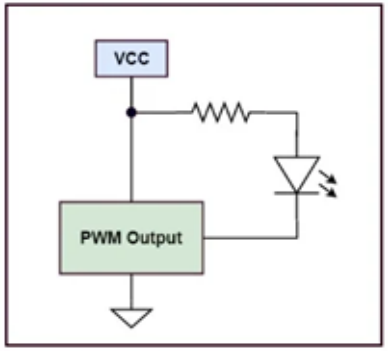


Duty Cycle = (Compare Value / Max Value) X 100%

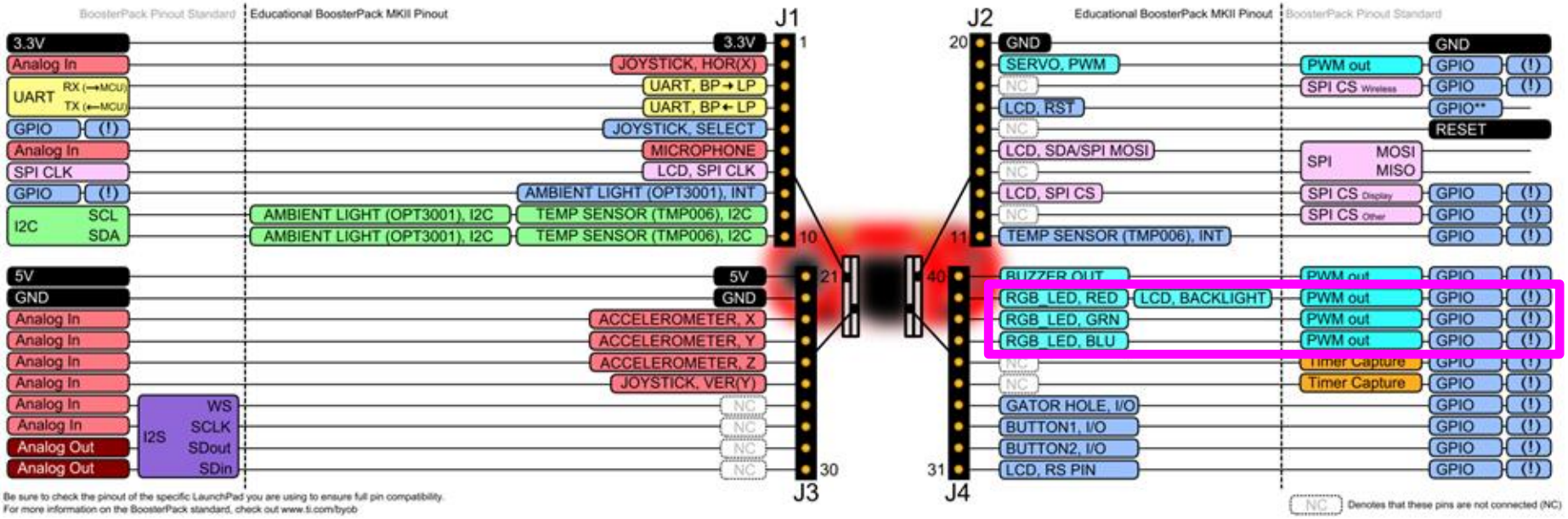
Controlling LED Intensity with a PWM Timer

- LED is on when the timer output is low and off when the output is high.
- The timer creates a **rapidly changing output**
  - turns the LED on and off fast enough
  - **the eye averages** the “on” time and perceives a change in intensity





PinOut Diagram



In PWM Mode

- You **do not need to** **manually toggle** the GPIO pin by processing in the ISR:
  - In PWM mode, the GPIO pin is **automatically toggled**.
- Pins are internally connected to the timer peripheral.
  - Internally, capture-compare registers are used to adjust the duty cycle.

Controlling LED Intensity with a PWM Timer

- Pins 5.6, 2.4 and 2.6 for RGB

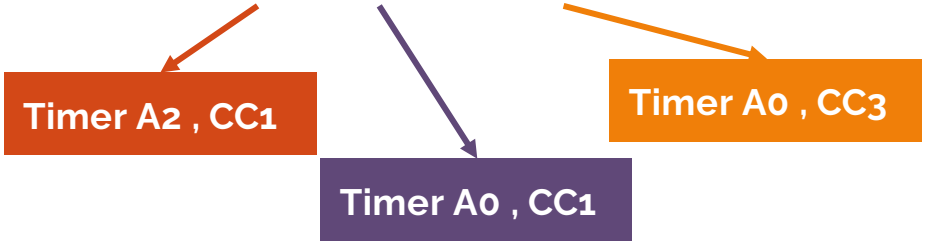


Table 6-75. Port P5 (P5.6 and P5.7) Pin Functions

PIN NAME (P5.x)	x	FUNCTION	CONTROL BITS OR SIGNALS[1]		
			P5DIR.x	P5SEL1.x	P5SEL0.x
P5.6/TA2.1/VREF+/VeREF+/C1.7	6	P5.6 (I/O)	I: 0; O: 1	0	0
		TA2.CCI1A	0	0	1
		TA2.1	1		
		N/A	0	1	0
		DVSS	1		
		VREF+, VeREF+, C1.7[2][3]	X	1	1
P5.7/TA2.2/VREF-/VeREF-/C1.6	7	P5.7 (I/O)	I: 0; O: 1	0	0
		TA2.CCI2A	0	0	1
		TA2.2	1		
		N/A	0	1	0
		DVSS	1		
		VREF-, VeREF-, C1.6[2][3]	X	1	1

Table 6-67. Port P2 (P2.4 to P2.7) Pin Functions

PIN NAME (P2.x)	x	FUNCTION	CONTROL BITS OR SIGNALS[1]			
			P2DIR.x	P2SEL1.x	P2SEL0.x	P2MAPx
P2.4/PM_TA0.1[2]	4	P2.4 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI1A	0	0	1	default
		TA0.1	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.5/PM_TA0.2[2]	5	P2.5 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI2A	0	0	1	default
		TA0.2	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.6/PM_TA0.3[2]	6	P2.6 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI3A	0	0	1	default
		TA0.3	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.7/PM_TA0.4[2]	7	P2.7 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI4A	0	0	1	default
		TA0.4	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			

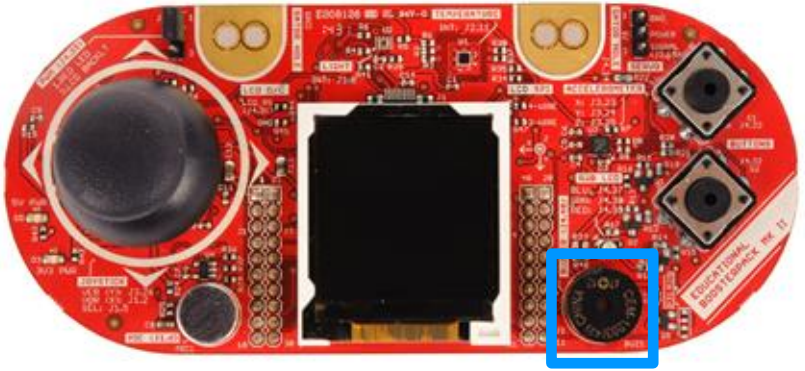
Lab#3:

- Use buttons to increase/decrease the intensity of the LEDs.

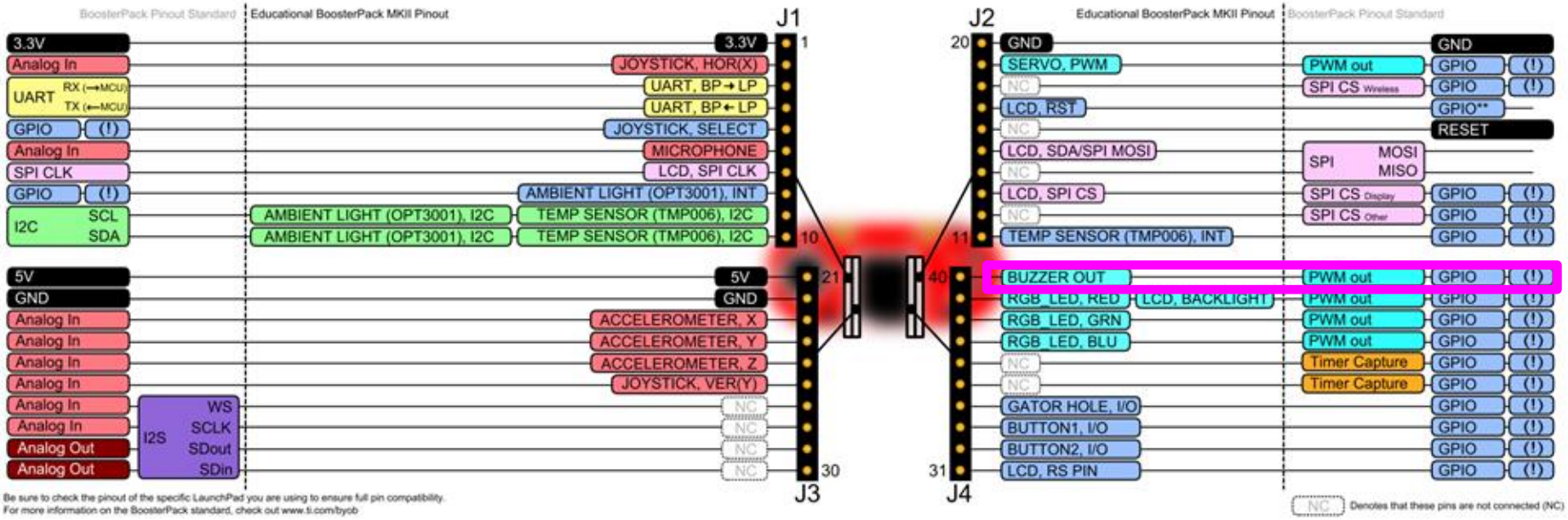
Piezo Buzzer

- The CUI CEM-1203(42) **piezo buzzer** can play various frequencies based on the user-provided PWM signal.
  - You can even play different tones back to back to create a song.

BoosterPack Plug-in Module Header Connection	Pin Function
J4.40	Buzzer input



PinOut Diagram



Piezo Buzzer

- P2.7 is the buzzer
  - Set timer mode and adjust PWM

Table 6-67. Port P2 (P2.4 to P2.7) Pin Functions

PIN NAME (P2.x)	x	FUNCTION	CONTROL BITS OR SIGNALS[1]			
			P2DIR.x	P2SEL1.x	P2SEL0.x	P2MAPx
P2.4/PM_TA0.1[2]	4	P2.4 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI1A	0	0	1	default
		TA0.1	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.5/PM_TA0.2[2]	5	P2.5 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI2A	0	0	1	default
		TA0.2	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.6/PM_TA0.3[2]	6	P2.6 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI3A	0	0	1	default
		TA0.3	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			
P2.7/PM_TA0.4[2]	7	P2.7 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI4A	0	0	1	default
		TA0.4	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			

Timer A0, CC4

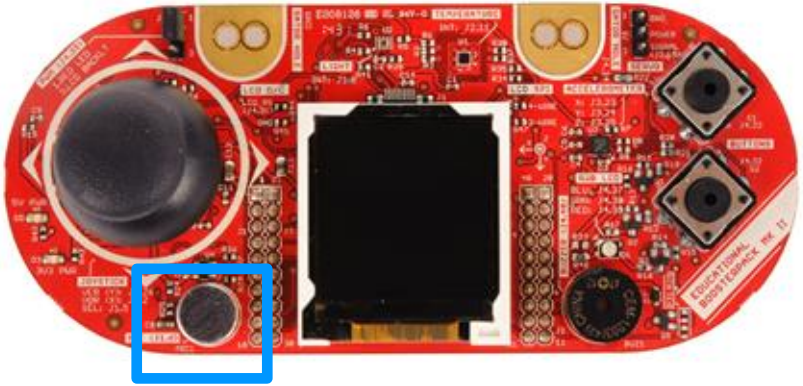
Lab #4:

- Try to produce some sounds via the buzzer
  - Buzzer will emit a sound according to the frequency and amplitude of vibration.
  - It allows to play notes and to recreate simple melodies.
- What do we need to do to play music with Buzzer?

Microphone

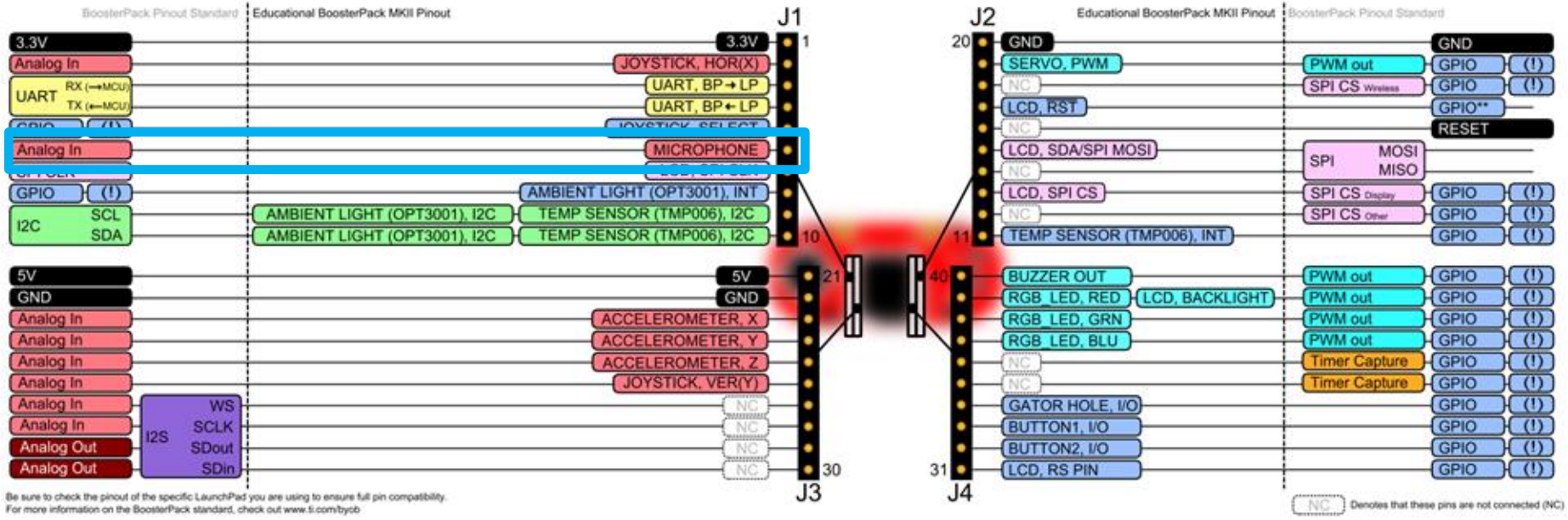
- The CUI CMA-4544PF-W **microphone** uses an OPA344 operational amplifier to boost the output of the microphone.
  - The human ear can hear frequencies between 0 and 20 kHz and the operating range of the microphone is 20 Hz to 20 kHz.

BoosterPack Plug-in Module Header Connection	Pin Function
J1.6	Microphone Output



PinOut Diagram

- Pin 4.3 on launchpad

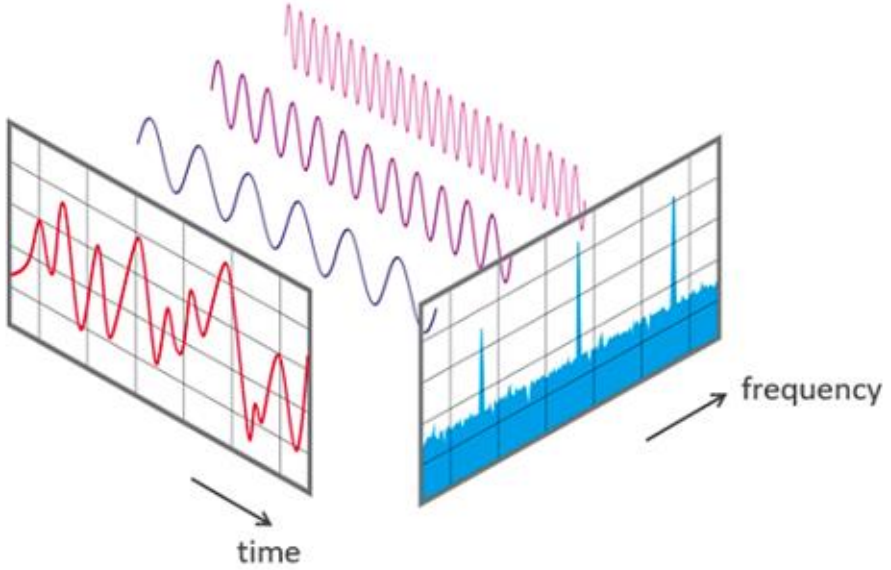


Lab #5: Detect Noise!

- Write a program that will blink an LED in a noisy environment.

FFT (Fast Fourier Transform)

- A transform takes one signal and turns it into another signal
  - From time domain to the frequency domain
- More details: https://www.youtube.com/watch?v=h7apO7q16V0&ab_channel=Reducible



Lab #6: Check FFT Example with the Microphone.

- How can we use the frequencies to understand activities?

Energia

- https://energia.nu/
  - Brings the Wiring and Arduino framework to the Texas Instruments MSP432 LaunchPad.
- You can check and see if it works
  - Feel free to use it too.

